Power BI DAX Functions

Power BI DAX Functions

From quick sums to time intelligence and filter logic

61 Essential Functions

DAX Measure

```
Sales YoY =
DIVIDE([Sales] - [Sales LY],
[Sales LY])
```

+12.4%

By

ADITYA KALYAN

DATA ENGINEER



Ebook

01 Aggregation & Math

`SUM()`	Adds all values in a column
`AVERAGE()`	Mean of all values in a column
`MIN()`	Smallest value in a column
`MAX()`	Largest value in a column
`COUNT()`	Counts numeric values
`COUNTROWS()`	Counts the rows in a table
`DISTINCTCOUNT()`	Counts unique values
`DIVIDE()`	Safe division that handles zero

```
Total Sales = SUM(Sales[Amount])

Orders = DISTINCTCOUNT(Sales[OrderID])

Avg Order = DIVIDE([Total Sales], [Orders])
```

Total Sales	1.24M
Orders	3,860
Avg Order	321

02 Logical

`IF()`	Returns a value based on a test
`SWITCH()`	Cleaner alternative to nested IFs
`AND()`	TRUE when both conditions are met
`OR()`	TRUE when any condition is met
`NOT()`	Reverses TRUE and FALSE
`IFERROR()`	Returns a fallback if there is an error
`COALESCE()`	Returns the first non-blank value

```
Status =

  IF([Sales] >= [Target],

     "On Track", "Behind")
```

Sales 120K | Target 100K

On Track

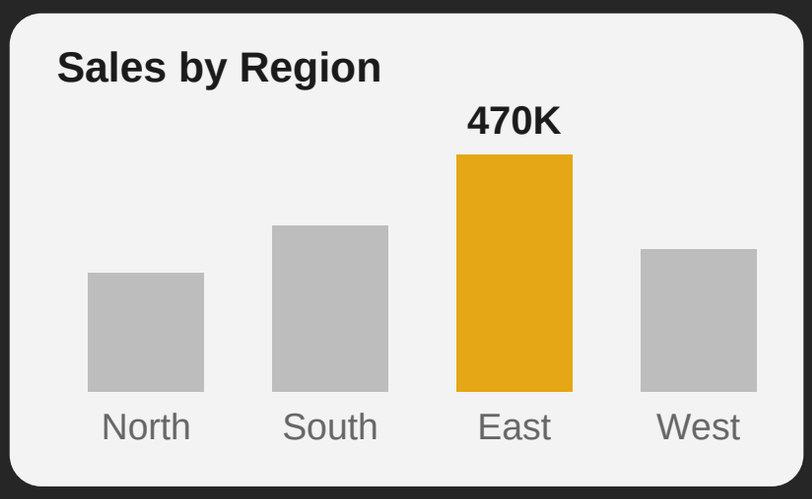
Sales 80K | Target 100K

Behind

03 Filter & Context

`CALCULATE()`	Evaluates an expression with new filters
`FILTER()`	Returns the rows that meet a condition
`ALL()`	Removes filters from a table or column
`ALLEXCEPT()`	Removes filters except chosen columns
`REMOVEFILTERS()`	Clears filters inside CALCULATE
`VALUES()`	Unique values in the current filter
`SELECTEDVALUE()`	Returns the single selected value
`KEEPFILTERS()`	Adds a filter without replacing others

```
East Sales =
CALCULATE(
  SUM(Sales[Amount]),
  Sales[Region] = "East"
)
```



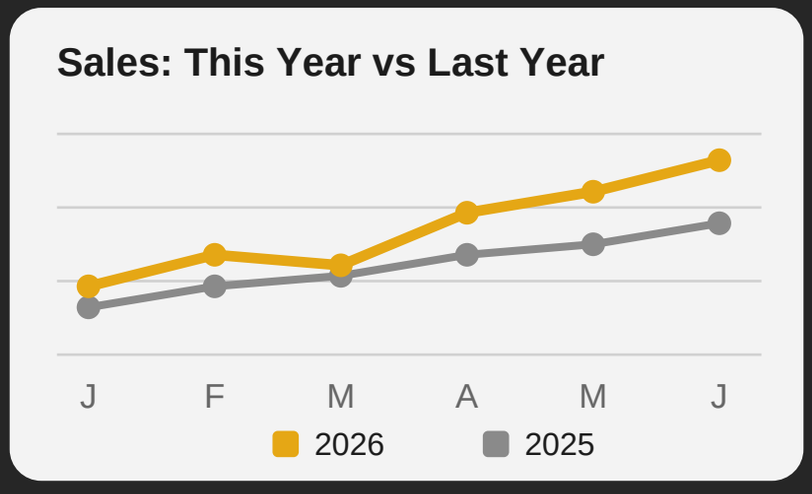
04 Time Intelligence

`TOTALYTD()`	Year-to-date total
`TOTALMTD()`	Month-to-date total
`TOTALQTD()`	Quarter-to-date total
`SAMEPERIODLASTYEAR()`	Same dates in the previous year
`DATEADD()`	Shifts dates by days, months or years
`PARALLELPERIOD()`	Shifts a full period forward or back
`DATESBETWEEN()`	Dates between a start and an end

```
Sales LY =

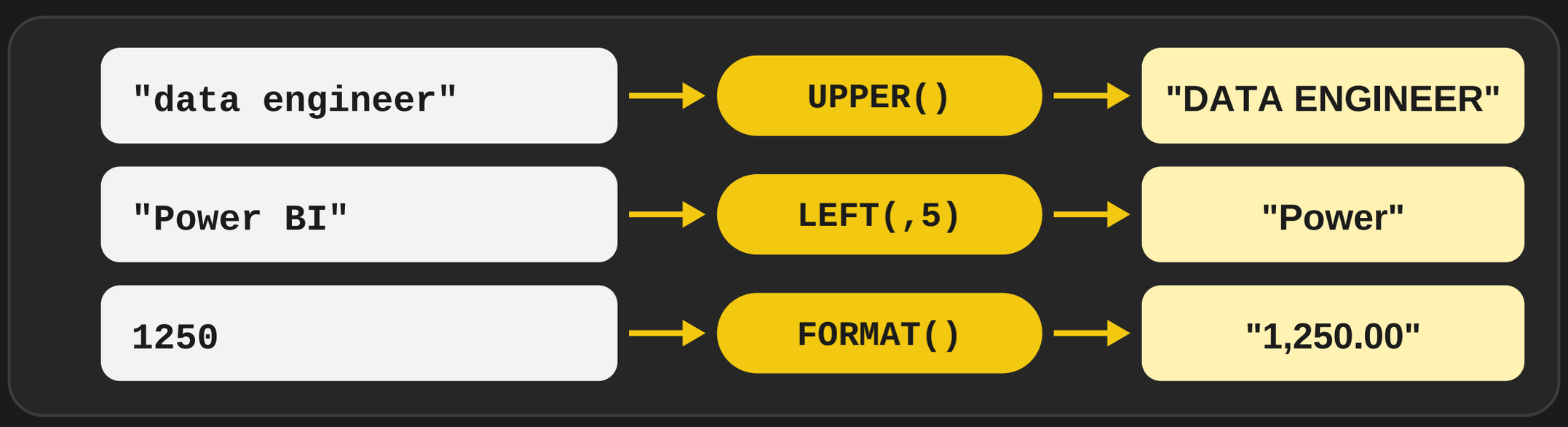
CALCULATE([Sales],

  SAMEPERIODLASTYEAR('Date'[Date]))
```



05 Text

`LEFT()`	First characters of a text
`RIGHT()`	Last characters of a text
`MID()`	Characters from the middle
`LEN()`	Number of characters
`UPPER()`	Converts text to capitals
`TRIM()`	Removes extra spaces
`FORMAT()`	Converts a value to formatted text
`CONCATENATEX()`	Joins values from a table



06 Date & Time

`TODAY()`	Current date
`NOW()`	Current date and time
`DATE()`	Builds a date from year, month, day
`YEAR()`	Year number of a date
`MONTH()`	Month number of a date
`EOMONTH()`	Last day of a month
`DATEDIFF()`	Difference between two dates
`CALENDAR()`	Creates a table of dates

```
Date =

CALENDAR(

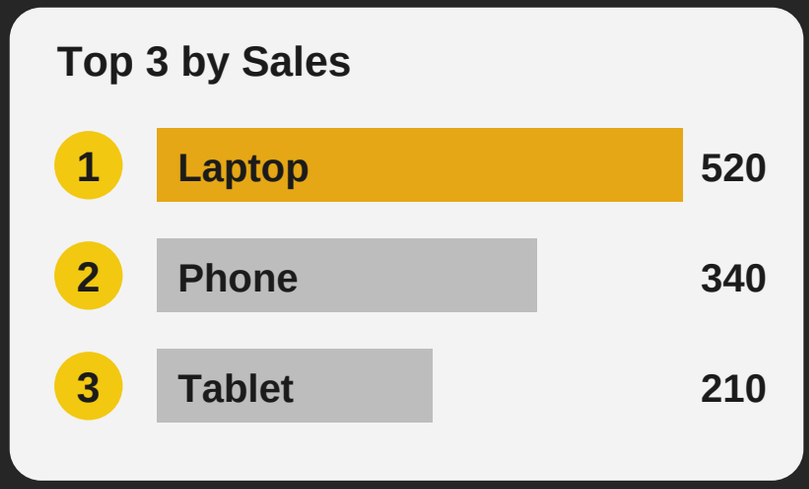
  DATE(2026,1,1), DATE(2026,12,31))
```

Date	Year	Month	Day
01-Jan-2026	2026	Jan	Thu
02-Jan-2026	2026	Jan	Fri
03-Jan-2026	2026	Jan	Sat
04-Jan-2026	2026	Jan	Sun

07 Iterators & Tables

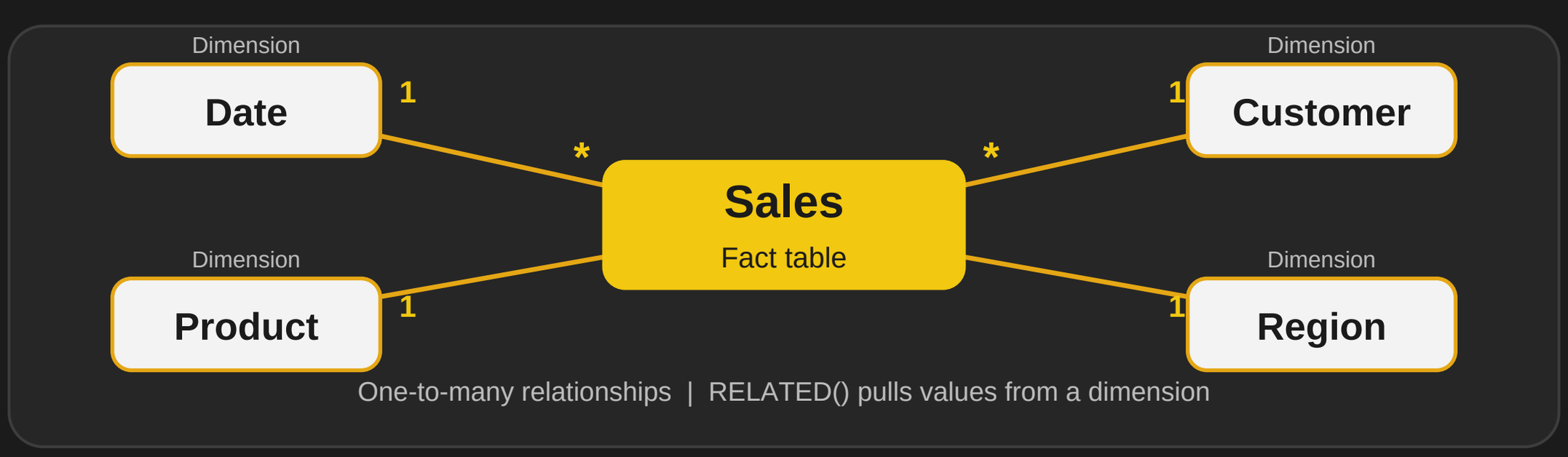
`SUMX()`	Sums an expression row by row
`AVERAGEX()`	Averages an expression row by row
`MAXX()`	Largest result of a row expression
`RANKX()`	Ranks values in a table
`TOPN()`	Returns the top N rows
`ADDCOLUMNS()`	Adds calculated columns to a table
`SUMMARIZE()`	Groups a table with totals

```
Top 3 Products =
TOPN(3,
  VALUES(Products[Name]),
  [Sales])
```



08 Relationships & Info

`RELATED()`	Fetches a value from the one side
`RELATEDTABLE()`	Returns related rows from the many side
`LOOKUPVALUE()`	Finds a value by matching columns
`USERELATIONSHIP()`	Activates an inactive relationship
`CROSSFILTER()`	Changes the filter direction
`ISBLANK()`	Checks for a blank value
`HASONEVALUE()`	TRUE when one value is in context
`ISFILTERED()`	TRUE when a column is filtered



One-to-many relationships | RELATED() pulls values from a dimension

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Thank you

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ADITYA KALYAN

DATA ENGINEER

